Curriculum-Enhanced Reinforcement Learning for Robust Humanoid Locomotion

Yi Zhou, Jianbin Qiu, *Fellow, IEEE*, Shixiang Jia, Fenglei Ni, and Wei Zhang, *Member, IEEE*

***Abstract*—The control of humanoid locomotion remains one of the most formidable challenges in robotics. Conventional model-based approaches not only rely extensively on manual design but also exhibit limited generalization across diverse tasks and environments. To overcome these limitations, a curriculum-enhanced reinforcement learning framework is proposed in this work for training robust locomotion. Given the strong temporal dependencies inherent in humanoid locomotion, Mamba-2 is employed as the backbone of the Actor-Critic network, enabling efficient temporal modeling of historical observations and content-based reasoning with linear spatiotemporal complexity. Furthermore, to eliminate the tendency of humanoid robots to favor low-speed motion in order to maintain gait stability and balance, which often leads to inaccurate velocity command tracking, a command-guided curriculum learning (CGCL) method is proposed to improve responsiveness to velocity commands. Experimental results demonstrate that the proposed Mamba-based framework achieves state-of-the-art (SOTA) performance, while CGCL significantly enhances the accuracy of velocity command tracking. Moreover, sim-to-real transfer experiments confirm both the robustness and the seamless deployability of the proposed framework on physical humanoid platforms.**

***Note to Practitioners*—Humanoid robots offer superior adaptability and performance in human-centered, unstructured environments, but controlling their whole-body locomotion remains challenging. Most existing methods rely on model-based designs that require extensive manual tuning, limiting their flexibility. This work proposes a novel curriculum-enhanced reinforcement learning framework that improves robustness and generalization for humanoid locomotion. On one hand, the framework utilizes Mamba-2 as the backbone of the Actor-Critic architecture, improving computational efficiency compared with current Transformer-based methods while retaining the ability to reason over complex movement sequences. On the other hand, a command-guided curriculum learning (CGCL) method is proposed to enhance the ability of the robot to accurately track velocity commands. Experimental results demonstrate that the proposed method exhibits strong robustness and precise command-tracking capabilities, highlighting its significant potential for real-world applications in humanoid locomotion.**

***Index Terms*—Reinforcement learning, learning-based control, end-to-end control, curriculum learning, humanoid locomotion.**

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I. Introduction

HUMANOID robots, owing to their anthropomorphic morphology, possess a unique ability to operate in human-centric environments, making them one of the most promising embodiments for realizing general embodied intelligence. Nevertheless, the intrinsic properties of underactuation, pronounced instability, high dimensionality, and nonlinear dynamics render the control of humanoid locomotion among the most formidable challenges in the field of robotics [1], [2]. Conventional approaches, largely inspired by human walking, have relied on inverted pendulum models [3], zero-moment point (ZMP) [4], and model predictive control (MPC) [5]. While these methods emphasize capturing the system dynamics through precise mathematical formulations, the resulting controllers are typically relying on partial or complete manual design [6]. Such reliance on manual design and frequent reconfiguration severely limits their scalability and generalization to diverse environments and tasks.

To overcome the inherent limitations of model-based control, learning-based approaches have emerged, primarily taking the form of imitation learning (IL) and reinforcement learning (RL). In IL-based methods, humanoid robots acquire skills directly from expert demonstrations, making the learned policies heavily dependent on the quantity and quality of demonstrations [7], which consequently constrains their capacity to generalize. In contrast, RL-based methods enable humanoid robots to learn autonomously through continuous interaction with the environment, eliminating the reliance on expert demonstrations. By engaging in exploration and exploitation, RL-based approaches optimize policies that exhibit superior robustness and generalization [8]. In [9], and [10], RL is utilized to train humanoid locomotion policies, with particular emphasis on the design of reward functions tailored to promote stable and efficient movement. Moreover, temporal sequence models, including Long Short-Term Memory (LSTM) [11] and Temporal Convolutional Network

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(TCN) [12], are incorporated into the policy networks to address locomotion tasks characterized by pronounced temporal dependencies, thereby enhancing adaptability to dynamic environments. Furthermore, the adoption of causal Transformers as the policy networks facilitates content-based reasoning, providing a more expressive mechanism for modeling long-range temporal dependencies [13]. While Transformers have shown promise in this regard, recent advances in large language models (LLMs) and computer vision [14] indicate that the Mamba architecture [15], [16] offers even greater potential. By surpassing Transformer-based models in multiple instances [17], [18], Mamba has emerged as a compelling alternative. Inspired by these developments, the present work explores the adoption of Mamba as the backbone architecture for the end-to-end control of humanoid locomotion.

Experimental observations show that the policy trained with RL for humanoid locomotion exhibits a marked tendency to march in place or move at low speeds, failing to accurately follow velocity commands. This behavior arises because maintaining gait and balance is easier at lower speeds, which causes their associated rewards to dominate in multi-objective optimization and overshadow the velocity tracking term [19]. Simply increasing the weight of the velocity tracking reward is not an effective solution, as the policy is typically unstable in the early training phase, and excessive emphasis on velocity tracking can disrupt gait stability and balance, potentially leading to policy failure [20]. To address these challenges, command-guided curriculum learning (CGCL) is proposed in this work, which constructs a curriculum specifically for velocity commands and employs tracking rewards to form closed-loop feedback, progressively enhancing the ability of the policy to achieve precise velocity tracking.

Motivated by the above discussion, a curriculum-enhanced hierarchical RL framework for the end-to-end control of humanoid locomotion is proposed in this work, which demonstrates strong robustness and generalization capabilities. The main contributions of this paper can be summarized as follows:

1) The Mamba model is employed as the backbone network to strengthen the temporal modeling capacity of the policy. To the best of our knowledge, this is the first study to apply Mamba to the control of humanoid locomotion.
2) A command-guided curriculum learning (CGCL) method is proposed, enhancing the accurate tracking and execution of velocity commands by humanoid robots.
3) Comparative experimental results demonstrate that the proposed method achieves state-of-the-art (SOTA) performance, and sim-to-real evaluations confirm seamless deployment on real humanoid robots.

The remainder of this paper is organized as follows. Section II formulates the problem and provides an overview of the Mamba architecture. Section III details the proposed method, including policy training, reward design, and the CGCL method. Section IV presents the results of comparative studies, ablation studies, and sim-to-real evaluations. Finally, Section V concludes the paper.

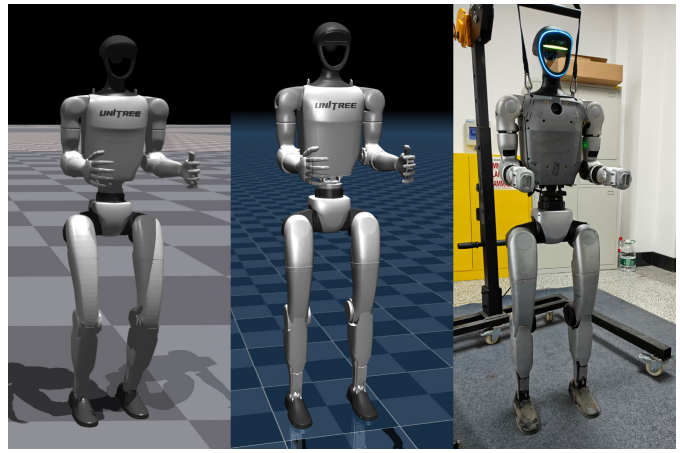

Fig. 1. The Unitree-G1 humanoid robot in Isaac Gym, MuJoCo, and real-world physical environments.

II. Problem Formulation and Preliminaries

A. Reinforcement Learning for Humanoid Locomotion

The end-to-end control of humanoid locomotion can be formalized as a Partially Observable Markov Decision Process (POMDP) [21], which is typically represented by a six-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, r, \gamma)$. Here, $\mathcal{S}$ denotes the set of true underlying system states that are not directly observable, $\mathcal{A}$ represents the set of possible actions that the agent can execute to influence the system, and $\mathcal{O}$ is the set of observations providing partial and noisy information about the true states. $\mathcal{P}(s_{t+1}|s_t, a_t)$ defines the state transition probability, specifying the likelihood of transitioning to state s_{t+1} given the current state s_t and action a_t. $r(s_t, a_t)$ denotes the reward function, which assigns a scalar value to quantify the immediate feedback resulting from the transition. $\gamma \in [0, 1]$ represents the discount factor that controls the trade-off between immediate and future rewards. In deep reinforcement learning (DRL), $\pi_\theta(a_t|s_t)$ is a neural network-based policy that maps the state s_t to an action a_t, where θ represents the learnable parameters of the neural network. The objective of RL is to optimize the policy $\pi_\theta(a_t|s_t)$ to maximize the expected return [22], defined as the cumulative discounted reward over an entire episode:

$$\mathcal{J}(\pi_\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t r(s_t, a_t)\right], \tag{1}$$

where $\mathbb{E}_{\pi_\theta}$ denotes the expectation taken with respect to the trajectory distribution induced by the policy π_θ, and T denotes the episode horizon.

In this study, the Unitree-G1 humanoid robot is adopted as the experimental platform, as illustrated in Fig 1. Throughout both training and deployment, the policy receives only proprioceptive information as input. To improve the learning efficiency, the base linear velocity, which is accessible exclusively within the simulation environment, is incorporated into the input of the critic network as privileged information. Given the inherently periodic nature of hip, knee, and ankle joint movements in bipedal locomotion, the phase of the gait cycle

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is encoded using a pair of sinusoidal signals, expressed as:

$$\text{Periodic Gait}(t) = \left[sin\left(\frac{2\pi t}{T_c}\right), cos\left(\frac{2\pi t}{T_c}\right)\right], \tag{2}$$

where T_c denotes the duration of a complete gait cycle. Furthermore, to reduce the complexity of the control problem, the upper-limb degrees of freedom are fixed, and the output actions correspond exclusively to the target positions of the 12 lower-limb joints. A detailed specification of the observation and privileged observation spaces is presented in Table I. Compared to standard observations, privileged observations incorporate quantities that cannot be accurately measured by onboard sensors in the real world, specifically base velocities, which are available only in simulation.

TABLE I
SUMMARY OF OBSERVATION AND PRIVILEGED OBSERVATION SPACES

Components	Dims	Observation	Privileged Observation
Angular Velocity	3	✓	✓
Projected Gravity	3	✓	✓
Commands	3	✓	✓
Joint Position	12	✓	✓
Joint Velocity	12	✓	✓
Last Action	12	✓	✓
Periodic Gait	2	✓	✓
Base Velocity	3		✓

B. Mamba Preliminaries

State space models (SSMs), recognized for their efficiency and effectiveness in sequence modeling, are rapidly emerging as a prominent research focus. By embedding discrete SSMs within neural network architectures, researchers have developed Mamba and a range of related variants. Compared to SSMs in classical control theory, which are utilized to describe system dynamic characteristics, neural SSMs in deep learning can be understood as sequence-to-sequence mappings where all parameter matrices are learnable.

As continuous-time latent state models, SSMs map a one-dimensional input $x(t)$ to an N-dimensional latent state $h(t)$ before projecting to a one-dimensional output $y(t)$, which can be expressed as:

$$\begin{cases} \dot{h}(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{cases} \tag{3}$$

where the state transition matrix $A \in \mathbb{R}^{N\times N}$ governs the dynamics of the latent state $h(t)$, the input matrix $B \in \mathbb{R}^{N\times 1}$ encodes the effect of the input $x(t)$, and the output matrix $C \in \mathbb{R}^{1\times N}$ projects the latent state $h(t)$ to the output $y(t)$. To enable the application of the continuous-time formulation (3) in sequence modeling, discretization must be performed using the Zero-Order Hold (ZOH) method. The resulting discrete-time SSMs and the corresponding discretization rule [23] can be represented as:

$$\begin{cases} h_k &= \bar{A}h_{k-1} + \bar{B}x_k \\ y_k &= \bar{C}h_k \\ \bar{A} &= \mathrm{e}^{T_s A} \\ \bar{B} &= (T_s A)^{-1}(\mathrm{e}^{T_s A} - I)T_s B \\ \bar{C} &= C \end{cases} \tag{4}$$

where T_s denotes the sampling period, which influences both the numerical stability and the dynamic performance of the discretized model. As shown in (4), the discrete-time SSMs can be interpreted as a sequence-to-sequence mapping from x_k to y_k [23]. The essence of applying SSMs to sequence modeling lies in encapsulating this process into a unified module, where the matrices A, B, and C are treated as learnable parameters. However, due to their inherently sequential structure, the aforementioned recurrent SSMs are unable to be parallelized efficiently on graphics processing units (GPUs), resulting in suboptimal training efficiency. By leveraging the linear time-invariant (LTI) property of the models and assuming a zero initial state $h_{-1} = 0$, the state variable h_k can be recursively eliminated, as follows:

$$y_k = \bar{C}(\bar{A})^k \bar{B}x_0 + \bar{C}(\bar{A})^{k-1}\bar{B}x_1 + \ldots + \bar{C}\bar{B}x_k. \tag{5}$$

Furthermore, (5) can be rewritten as a global convolution:

$$\begin{cases} y &= \bar{K} * x \\ \bar{K} &= (\bar{C}\bar{B}, \bar{C}\bar{A}\bar{B}, \ldots, \bar{C}(\bar{A})^{L-1}\bar{B}) \end{cases} \tag{6}$$

where $\bar{K}$ is referred to as the convolution kernel of SSMs, and L denotes the length of the input sequence. The convolutional form of discrete-time SSMs (6) enables efficient parallel computation and is therefore typically employed for training, whereas the recurrent form (4) is more suitable for autoregressive inference with explicit stepwise dependencies.

Nevertheless, a fundamental limitation of directly applying SSMs to sequence modeling is their inability to perform content-based reasoning, which refers to the capability to discriminately focus on informative elements within a sequence while disregarding irrelevant information. To address this limitation, researchers have introduced a mechanism that transforms the parameters of the discrete-time SSMs as functions of the input. This mechanism allows the SSMs to dynamically propagate or forget information along the sequence length dimension based on the current input token, forming the core design of Mamba [15], which can be expressed as:

$$\begin{cases} h_k &= \bar{A}_k h_{k-1} + \bar{B}_k x_k \\ y_k &= \bar{C}_k h_k \\ \bar{A}_k &= \mathrm{e}^{\Delta_k A} \\ \bar{B}_k &= (\Delta_k A)^{-1}(\mathrm{e}^{\Delta_k A} - I)\Delta_k B_k \\ \bar{C}_k &= C_k \\ B_k &= \texttt{Linear}_N(x_k) \\ C_k &= \texttt{Linear}_N(x_k) \\ \Delta_k &= \texttt{Softplus}(\text{Broadcast}_D(\texttt{Linear}_1(x_k)) + b_k) \end{cases} \tag{7}$$

where $\texttt{Linear}_d(x_k)$ denotes a learnable linear transformation that projects the last dimension of x_k to d dimensions, the Softplus function is defined as $\texttt{Softplus}(u) = \ln(1+\mathrm{e}^u)$, and b_k represents a trainable parameter with D dimensions. Within the selective SSMs, Δ_k serves as the most critical element, functioning analogously to the gating mechanism in RNNs by modulating the selectivity for the input x_k. Moreover, imposing

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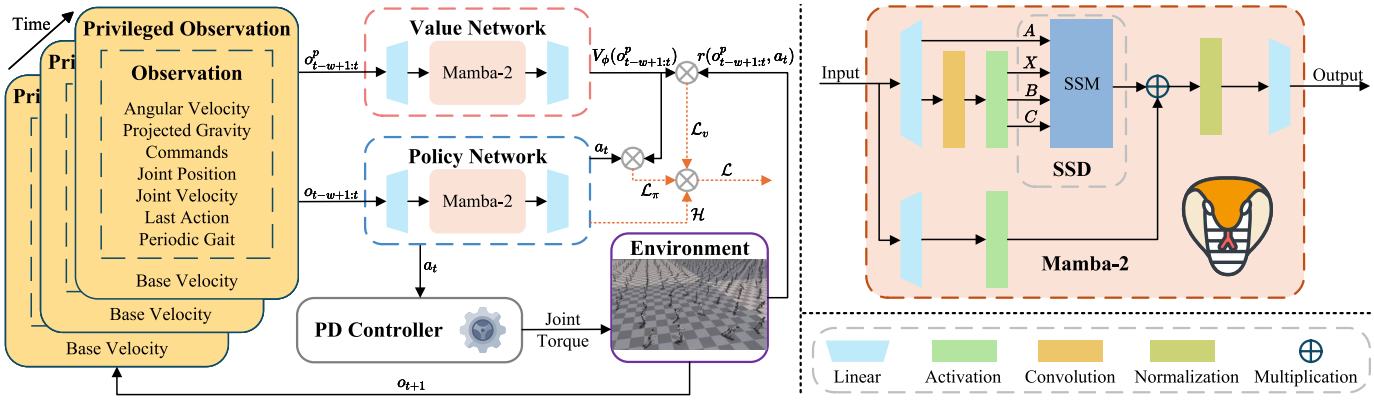

Fig. 2. Hierarchical policy architecture, consisting of a high-level Mamba-based Actor-Critic framework and a low-level PD controller.

selectivity on B_k and C_k allows the model to decide whether to pass the current input x_k into the state h_k and whether to allow the current state h_k to produce the output y_k. Although A could in principle exhibit input-dependent selectivity, (7) indicates that the selectivity of Δ_k already guarantees that of $\bar{A}_k$ and $\bar{B}_k$, so A is generally treated as a trainable parameter without selectivity for simplicity. The structure of the A matrix, which is crucial for controlling temporal dynamics, is initialized following S4D-Real [24] as a diagonal matrix with $A_{ii} = -(i+1)$. This standard initialization for sequence modeling is particularly suitable for locomotion tasks, as it effectively captures the temporal structures within historical observation sequences characterized by significant periodicity. This selective SSMs form the core design of Mamba, functioning as a mapping $X \mapsto Y$ in which the parameter matrices $\bar{A}$, $\bar{B}$, and $\bar{C}$ depend on the input X. Building on this, Mamba-2 [16] proposes the Structured State Space Duality (SSD) framework to further enhance parallelism, which can be interpreted as a mapping $\{\bar{A}, \bar{B}, \bar{C}, X\} \mapsto Y$, where the parameter matrices $\bar{A}$, $\bar{B}$ and $\bar{C}$, together with the input X, are generated simultaneously. In addition, compared to Mamba, Mamba-2 incorporates a normalization layer before the output linear layer to enhance training stability.

III. Methodology

A. Hierarchical Policy

In this work, a hierarchical policy architecture is employed to train robust locomotion for humanoid robots, structured as a high-level planner and a low-level controller. The high-level planner, operating at a frequency of 100 Hz, is responsible for generating desired joint positions for the upcoming planning horizon conditioned on proprioceptive observations. Specifically, these desired joint positions are produced in an end-to-end manner by the high-level planner, corresponding directly to the actions output by the Mamba-based policy network. The low-level controller, running at a higher frequency of 1000 Hz, utilizes a proportional-derivative (PD) control scheme to accurately track the desired joint positions provided by the high-level planner. In light of the real-time requirements, the low-level controller is expected to rapidly and accurately track the time-varying desired joint positions, taking the desired positions of each joint as input and producing the corresponding joint torques as output. This hierarchical policy architecture, as illustrated in Fig 2, enables the high-level planner to focus on policy reasoning, while the low-level controller guarantees precise execution by compensating for mechanical delays and disturbances within the inner control loop [25].

In the proposed Mamba-based policy, the high-level planner leverages the Asymmetric Actor-Critic architecture, which accelerates training and enhances the generalization capability of the learned policy. The fundamental principle of the Asymmetric Actor-Critic is to grant the value network access to privileged information available in the simulation environment during training, while restricting the policy network to partial observations that reflect the sensory inputs obtainable in the real world. The effectiveness of the asymmetric design stems from its ability to mitigate errors caused by state aliasing. In scenarios where the agent cannot accurately infer the true state due to partial observability, conventional Symmetric Actor-Critic architectures introduce estimation errors [26]. By enabling the value network to utilize complete observation information during training, the Asymmetric Actor-Critic architecture effectively eliminates this source of error, thereby improving the fidelity of value estimation and the robustness of the learned policy [27]. The policy network is optimized using the Proximal Policy Optimization (PPO) objective. PPO mitigates training instability by constraining the magnitude of policy updates, thereby preventing abrupt changes in the policy during learning, with its objective function defined as:

$$\mathcal{L}_\pi = \min\left[\frac{\pi_\theta(a_t|o_{t-w+1:t})}{\pi_{\theta'}(a_t|o_{t-w+1:t})} A^{\pi_{\theta'}}(o^p_{t-w+1:t}, a_t),\right.$$
$$\left.\text{clip}\left(\frac{\pi_\theta(a_t|o_{t-w+1:t})}{\pi_{\theta'}(a_t|o_{t-w+1:t})}, c_1, c_2\right) A^{\pi_{\theta'}}(o^p_{t-w+1:t}, a_t)\right] \quad (8)$$

where π_θ denotes the new policy to be optimized, and $\pi_{\theta'}$ represents the old policy used for data sampling. The constants c_1 and c_2 define the lower and upper bounds for the PPO clipping mechanism, respectively. Unlike conventional Actor-Critic networks that rely solely on single-step observations, the proposed approach incorporates a sequence of past observations as policy inputs, where w specifies the length of the historical context window considered. Specifically, $o_{t-w+1:t}$ denotes the observations, which serve as the input to the policy network, while $o^p_{t-w+1:t}$ represents the privileged observations,

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which are used as the input to the value network. The advantage function $A^{\pi_{\theta'}}$ is estimated using Generalized Advantage Estimation (GAE) [28], which provides a trade-off between Monte Carlo (MC) methods and Temporal Difference (TD) learning. The GAE can be formulated as:

$$A^{\pi_{\theta'}}(o^p_{t-w+1:t}, a_t) = \sum_{l=0}^{\infty}(\gamma\lambda)^l \delta_{t+l}, \tag{9}$$

where the hyperparameter $\lambda \in [0, 1]$ regulates the trade-off between bias and variance. In practical implementation, the summation in GAE is truncated according to the sequence length L, which can be expressed as:

$$A^{\pi_{\theta'}}(o^p_{t-w+1:t}, a_t) = \sum_{l=0}^{L-t-1}(\gamma\lambda)^l \delta_{t+l}, \tag{10}$$

where δ_t denotes the TD error, defined as:

$$\begin{aligned}\delta_t = {} & r(o^p_{t-w+1:t}, a_t) + \gamma V_\phi(o^p_{(t-w+1)+1:t+1}) \\ & - V_\phi(o^p_{t-w+1:t}),\end{aligned} \tag{11}$$

where $r(o_{t-w+1:t}, a_t)$ represents the immediate reward conditioned on the current observation window $o_{t-w+1:t}$ and action a_t. The value function $V_\phi(o^p_{t-w+1:t})$, parameterized by ϕ, estimates the expected return based on the given observation sequence $o^p_{t-w+1:t}$. The future value is estimated using the next observation window $o^p_{(t-w+1)+1:t+1}$, and discounted by a factor γ. At time step t, the GAE target return is defined as:

$$R_t = A^{\pi_{\theta'}}(o^p_{t-w+1:t}, a_t) + V_\phi(o^p_{t-w+1:t}). \tag{12}$$

The value network is trained to fit the cumulative discounted return using a squared error loss:

$$\mathcal{L}_v = \frac{1}{2}\left(V_\phi(o^p_{t-w+1:t}) - R_t\right)^2. \tag{13}$$

In addition, to encourage exploration and prevent the policy from prematurely converging to a deterministic solution, an entropy term is incorporated, which is defined as:

$$\mathcal{H}[\pi_\theta(\cdot|o_{t-w+1:t})] = -\mathbb{E}_{a_t\sim\pi_\theta}\left[\log \pi_\theta(a_t|o_{t-w+1:t})\right]. \tag{14}$$

A higher entropy value indicates greater uncertainty in the policy, leading to increased exploration by the agent. For humanoid robots operating in continuous action spaces, the policy is typically modeled as a Gaussian distribution, represented as:

$$\pi_\theta(a_t|o_{t-w+1:t}) = \mathcal{N}(\mu_\theta(o_{t-w+1:t}), \sigma^2_\theta(o_{t-w+1:t})), \tag{15}$$

where $\mu_\theta(o_{t-w+1:t})$ and $\sigma_\theta(o_{t-w+1:t}))$ denote the mean and standard deviation of the Gaussian distribution, respectively. At this point, the entropy value can be expressed as:

$$\mathcal{H}[\pi_\theta(\cdot|o_{t-w+1:t})] = \sum_{i=1}^{d_a}\left(\log(\sigma_\theta)_i + \frac{1}{2}\log(2\pi e)\right), \tag{16}$$

where d_a represents the dimensionality of the action space and the term $\frac{1}{2}\log(2\pi e)$ is a constant factor. Since the entropy of a Gaussian distribution is positively correlated with its standard deviation, maximizing this term prevents the action variance from vanishing rapidly. This mechanism prevents the policy from prematurely converging to a deterministic solution, thereby ensuring continuous exploration of the action space. Taking into account the policy loss, value loss, and entropy regularization term, a single optimizer is employed to jointly optimize both the policy and value networks. The overall loss function can be formulated as:

$$\mathcal{L} = -\mathcal{L}_\pi + \lambda_v \mathcal{L}_v - \lambda_{\mathcal{H}}\mathcal{H}, \tag{17}$$

where λ_v and $\lambda_{\mathcal{H}}$ are hyperparameters that control the relative contributions of the value loss and the entropy regularization term to the total loss. PPO implicitly constrains policy updates by clipping the objective function, thereby limiting abrupt changes in the policy. To more precisely quantify the extent of policy deviation, the Kullback-Leibler (KL) divergence between the new and old policies is measured, which can be expressed as:

$$d_{kl} = D_{\mathrm{KL}}(\pi_{\theta'}(\cdot|o_{t-w+1:t})\|\pi_\theta(\cdot|o_{t-w+1:t})). \tag{18}$$

Based on this measure, an adaptive learning rate adjustment strategy [29] is introduced, which can be expressed as:

$$\begin{cases}\alpha_{k+1} \leftarrow \max\left(10^{-5}, \dfrac{\alpha_k}{1.5}\right), & \text{if } d_{kl} > 2d^*_{kl} \\ \alpha_{k+1} \leftarrow \min(10^{-2}, 1.5\alpha_k), & \text{if } d_{kl} < 0.5d^*_{kl} \\ \alpha_{k+1} \leftarrow \alpha_k, & \text{otherwise.}\end{cases} \tag{19}$$

Here, α_k denotes the learning rate at the previous iteration, and α_{k+1} represents the learning rate for the current update. The term d^*_{kl} refers to the target KL divergence, serving as a reference setpoint that defines the desired magnitude of the deviation between the updated policy π_θ and the previous policy $\pi_{\theta'}$. This formulation enables an adaptive adjustment of the learning rate based on the observed KL divergence between consecutive policies, thereby ensuring that the magnitude of policy updates remains within a reasonable and controlled range.

B. Reward Design

In this work, the reward function is designed to guide the humanoid robot to accurately track user-specified velocity commands while simultaneously maintaining a stable and periodic gait characterized by natural footstep motion. To achieve these two overarching objectives, the scalar reward at each timestep is formulated as a combination of velocity tracking, gait adherence, and regularization terms. A comprehensive specification of each reward component is provided in Table II.

In Table II, only the scales associated with the linear and angular velocity tracking rewards are positive, indicating that the agent is encouraged to closely follow the velocity commands. In contrast, all other reward components are assigned negative scales, signifying that they are treated as penalties. Here, v_x, v_y, and v_z represent the linear velocities along the x, y, and z axes, respectively, while ω_x, ω_y, and ω_z denote the angular velocities about the x, y, and z axes. CMD_x, CMD_y, and CMD_z correspond to the velocity commands along the x and y directions and about the z-axis. The velocity tracking rewards incentivize the robot to accurately follow the velocity commands as closely as possible, while simultaneously minimizing the magnitudes of velocities in the other directions,

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TABLE II
REWARD FUNCTION DESIGN

Reward	Equation
Linear Velocity Tracking	$e^{-4\|\|v_{xy}-\mathrm{CMD}_{xy}\|\|^2}$
Angular Velocity Tracking	$e^{-4\|\|\omega_z-\mathrm{CMD}_z\|\|^2}$
Linear Velocity Penalty	$\|\|v_z\|\|^2$
Angular Velocity Penalty	$\|\|\omega_{xy}\|\|^2$
Base Orientation Tracking	$\|\|g_{xy}\|\|^2$
Base Height Tracking	$\|\|h-h^*\|\|^2$
Contact Phase Alignment	$\sum_{i=1}^{2}(\mathbb{I}(c_i)\oplus\mathbb{I}(p_i))$
Static Contact	$\sum_{i=1}^{2}[\mathbb{I}(c_i)\|\|v_i\|\|^2]$
Feet Swing Height	$\sum_{i=1}^{2}(1-\mathbb{I}(c_i))\,(f_i-0.08)^2$
Joint Position Limit	$\sum_{i=1}^{12}[\max(q_i-q_{max},0)-\min(q_i-q_{min},0)]$
Joint Velocity Limit	$\|\|\dot{q}\|\|^2$
Joint Acceleration Limit	$\|\|\ddot{q}\|\|^2$
Action Rate Limit	$\|\|a_t-a_{t-1}\|\|^2$

specifically v_z, ω_x, and ω_y. In addition, the velocity tracking rewards further incorporate terms for maintaining a flat base orientation and a desired base height. When the base is level, the projection of the gravity vector onto the x and y axes, denoted as g_{xy}, should be zero. The target height of the base h^* is set to 1 meter, and the base is urged to remain at this height during locomotion.

Gait adherence rewards are designed to align the actual gait pattern $\mathbb{I}(c_i)$ with the reference periodic gait $\mathbb{I}(p_i)$. These rewards also promote zero velocity at the feet during ground contact to reduce impact forces and slippage. Additionally, during non-contact phases, the feet are expected to maintain a lifted height of 0.08 meters to prevent spurious rewards caused by undesired ground contact or dragging.

Regularization terms primarily include soft constraints on joint positions q, as well as penalties on joint velocities $\dot{q}$, accelerations $\ddot{q}$ and action variations. These components suppress abrupt joint movements, prevent excessive torque generation, and ensure the smoothness of the policy outputs [30]. Collectively, all reward components jointly guide the humanoid robot to accurately follow the velocity commands while maintaining a stable, natural, and efficient gait.

C. Command-Guided Curriculum Learning

Due to the high dimensionality of the action space and the inherent difficulty of maintaining balance, end-to-end control of humanoid locomotion via DRL often suffers from low learning efficiency, training instability, and insufficient exploration during the initial stages. To mitigate these issues, curriculum learning has been recognized as an effective paradigm for facilitating policy optimization. By systematically organizing a progression of tasks from simple to complex, curriculum learning enables the agent to acquire the target skills more efficiently and robustly. Existing applications of curriculum learning for humanoid locomotion have primarily focused on terrain-based curricula, in which progressively more challenging terrains are introduced to allow the robot to gradually adapt to increasingly complex environments [29], [31].

Experimental observations reveal that, even with the inclusion of velocity tracking rewards, humanoid robots often resort to in-place stepping or low-speed walking rather than accurately following the velocity commands. This tendency can be attributed to the intrinsic characteristics of humanoid robots, including a high center of mass and a large number of articulated joints, which render them highly susceptible to balance loss and gait instability during high-speed locomotion. Consequently, the learned policy prioritizes stability over precise velocity tracking, resulting in conservative behaviors. Simply increasing the scales of the velocity tracking rewards is not an effective remedy, as the policy in the early stages of training has not yet converged to a reasonably optimal solution. Overemphasizing velocity tracking at this stage compromises gait stability and reduces robustness, and may even lead to policy collapse [32]. To address this challenge, the command-guided curriculum learning (CGCL) method is proposed as the core innovation of this work, which is formally illustrated in Algorithm 1. Unlike conventional curricula for humanoid locomotion that are primarily designed around terrain variations, CGCL constructs the curriculum based on velocity commands. By progressively increasing the velocity commands and incorporating feedback from velocity tracking rewards, CGCL ensures that the policy gradually acquires accurate command-following capability, ultimately yielding a robust policy capable of sustaining high-speed locomotion.

As shown in Algorithm 1, the essence of the proposed CGCL method lies in dynamically adjusting the velocity commands based on feedback from velocity tracking rewards, with its core design centered around three key components. First, the curricula for linear and angular velocity commands are designed separately due to their distinct physical properties and control challenges. Linear velocity primarily governs forward locomotion and overall stability, whereas angular velocity requires asymmetric coordination and is more sensitive to balance. Separate curricula ensure balanced skill acquisition and prevent the difficulty of one domain from hindering the other. Second, a dual-threshold detection mechanism is employed to evaluate the success or failure of the learning process. If the average reward exceeds the predefined high threshold τ_1, the learning at this stage is considered successful, and the corresponding velocity command is increased accordingly. If the average reward falls below the low threshold τ_2, a failure is recorded, incrementing the failure counters c_{fl} and c_{fa}. A failure state is only triggered when the cumulative number of failures reaches a preset limit c_t. This design provides a degree of fault tolerance, allowing the policy sufficient time to adapt and preventing premature adjustments caused by occasional failures. Finally, instead of directly decreasing the velocity commands upon entering a failure

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Algorithm 1 Command-Guided Curriculum Learning

Initialize:
Initialize the velocity commands $\mathrm{CMD}_x \leftarrow \pm 0.2$ m/s, $\mathrm{CMD}_y \leftarrow \pm 0.2$ m/s, $\mathrm{CMD}_z \leftarrow 0$ rad/s.
Initialize the upper limits of the velocity commands $\mathrm{CMD}_{xm} \leftarrow \pm 1$ m/s, $\mathrm{CMD}_{ym} \leftarrow \pm 1$ m/s, $\mathrm{CMD}_{zm} \leftarrow \pm 1$ rad/s.
Initialize the step size of the velocity command increment $\Delta_x \leftarrow 0.1$ m/s, $\Delta_y \leftarrow 0.1$ m/s, $\Delta_z \leftarrow 0.1$ rad/s.
Initialize the reward scales for linear velocity tracking $s_l \leftarrow s_{l0} \leftarrow 1.0$, and for angular velocity tracking $s_a \leftarrow s_{a0} \leftarrow 0.5$.
Initialize the high reward ratio threshold $\tau_1 \leftarrow 0.8$ and the low reward ratio threshold $\tau_2 \leftarrow 0.5$.
Initialize the failure count for linear velocity commands $c_{fl} \leftarrow 0$, and for angular velocity command $c_{fa} \leftarrow 0$.
Initialize the trigger count of the failure state $c_t \leftarrow 3$ and the increment factor of the reward scales $\beta \leftarrow 1.25$.
for each training episode **do**
  Execute the current velocity commands CMD_x, CMD_y, CMD_z.
  for each interaction episode **do**
    Interact for N steps and collect the velocity tracking rewards $r_{xy}^1, r_z^1, r_{xy}^2, r_z^2, \ldots, r_{xy}^N, r_z^N$.
  end for
  Compute the average velocity tracking rewards over N steps: $\bar{r}_{xy} = \frac{1}{N}\sum_{i=1}^{N} r_{xy}^i$, $\bar{r}_z = \frac{1}{N}\sum_{i=1}^{N} r_z^i$.
  /* Linear velocity command curriculum */
  if $|\mathrm{CMD}_x| \le |\mathrm{CMD}_{xm}|$ **and** $|\mathrm{CMD}_y| \le |\mathrm{CMD}_{ym}|$ **and** $\bar{r}_{xy} > \tau_1$ **then** ▹Exceed the high threshold
    $|\mathrm{CMD}_x| \leftarrow |\mathrm{CMD}_x| + \Delta_x$, $|\mathrm{CMD}_y| \leftarrow |\mathrm{CMD}_y| + \Delta_y$, $s_l \leftarrow s_{l0}$, $c_{fl} \leftarrow 0$.
  else if $|\mathrm{CMD}_x| \le |\mathrm{CMD}_{xm}|$ **and** $|\mathrm{CMD}_y| \le |\mathrm{CMD}_{ym}|$ **and** $\bar{r}_{xy} < \tau_2$ **then** ▹Below the low threshold
    $|\mathrm{CMD}_x| \leftarrow |\mathrm{CMD}_x|$, $|\mathrm{CMD}_y| \leftarrow |\mathrm{CMD}_y|$, $c_{fl} \leftarrow c_{fl} + 1$.
    if $c_{fl} == c_t$ **then** ▹Trigger the failure state and enlarge the reward scale
      $s_l \leftarrow \beta \times s_l$, $c_{fl} \leftarrow 0$
    end if
  else
    $|\mathrm{CMD}_x| \leftarrow |\mathrm{CMD}_x|$, $|\mathrm{CMD}_y| \leftarrow |\mathrm{CMD}_y|$.
  end if
  /* Angular velocity command curriculum */
  if $|\mathrm{CMD}_z| \le |\mathrm{CMD}_{zm}|$ **and** $\bar{r}_z > \tau_1$ **then** ▹Exceed the high threshold
    $|\mathrm{CMD}_z| \leftarrow |\mathrm{CMD}_z| + \Delta_z$, $s_a \leftarrow s_{a0}$, $c_{fa} \leftarrow 0$.
  else if $|\mathrm{CMD}_z| \le |\mathrm{CMD}_{zm}|$ **and** $\bar{r}_z < \tau_2$ **then** ▹Below the low threshold
    $|\mathrm{CMD}_z| \leftarrow |\mathrm{CMD}_z|$, $c_{fa} \leftarrow c_{fa} + 1$.
    if $c_{fa} == c_t$ **then** ▹Trigger the failure state and enlarge the reward scale
      $s_a \leftarrow \beta \times s_a$, $c_{fa} \leftarrow 0$
    end if
  else
    $|\mathrm{CMD}_z| \leftarrow |\mathrm{CMD}_z|$.
    end if
end for

state, the corresponding scaling factors s_l or s_a of the velocity tracking rewards are increased in this method. This mechanism preserves the full action exploration space while encouraging the policy to focus more effectively on velocity tracking. In addition, it avoids the oscillations and discontinuities caused by frequent switching between command magnitudes, thereby ensuring the stability and continuity of the learning process. Overall, these carefully designed mechanisms enable CGCL to guide the learning process progressively, improving both efficiency and robustness while maintaining stable and continuous behavior.

IV. Experiments and Results

To ensure reproducibility and facilitate fair comparison, the detailed hyperparameters employed in the training process and the network architecture specifications are summarized in Table III. All baseline methods utilize identical training configurations to ensure the validity of the comparative results.

A. Comparative Studies

The predominant network architectures employed in RL for humanoid locomotion are primarily based on multilayer perceptrons (MLPs) [10], LSTM [11], and Transformers [13]. To comprehensively assess the superiority of the proposed Mamba-based architecture, Mamba-2 is adopted as the backbone of the Actor-Critic network, and comparative experiments are conducted against these three representative baselines. All models are trained under identical conditions, including environment configuration, reward design, and hyperparameter settings. Furthermore, to ensure the fairness of the comparison, we adjusted the network configurations of each backbone so that all models possess a comparable number of learnable parameters. Specifically, for all these methods, the

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TABLE III
HYPERPARAMETERS AND NETWORK SPECIFICATIONS

Category	Parameter Name	Symbol	Value
Policy Parameters	Discount Factor	γ	0.99
	GAE Parameter	λ	0.95
	PPO Clip Lower Bound	c_1	0.8
	PPO Clip Upper Bound	c_2	1.2
	Value Loss Coefficient	λ_v	1.0
	Entropy Coefficient	$\lambda_{\mathcal{H}}$	0.01
	Target KL Divergence	d_{kl}^*	0.01
	Initial Learning Rate	α	10^{-3}
	Context Window Length	w	16
Mamba-2 Architecture	Hidden Dimension	d_{model}	256
	State Dimension	N	64
	Action Output Dimension	d_a	12
PD Controller Gains	Hip Stiffness	K_p^h	100
	Hip Damping	K_d^h	2
	Knee Stiffness	K_p^k	150
	Knee Damping	K_d^k	4
	Ankle Stiffness	K_p^a	40
	Ankle Damping	K_d^a	2

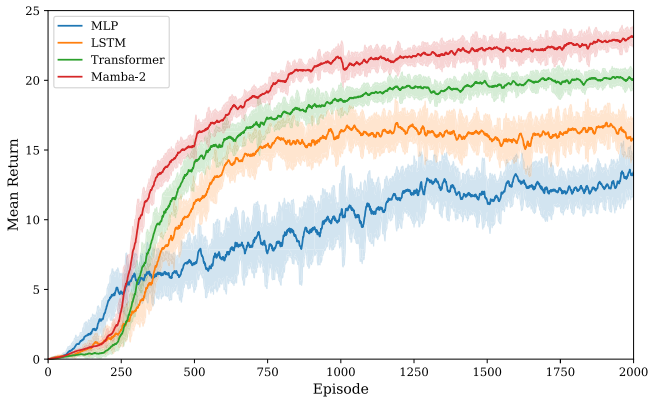

Fig. 3. Comparative experimental results demonstrate that the Mamba-based architecture surpasses the performance of three baselines.

velocity commands are sampled uniformly, with the forward linear velocity within $[-1, 1]$ m/s, the sideways linear velocity within $[-1, 1]$ m/s, and the turning angular velocity within $[-1, 1]$ rad/s.

The learning curves presented in Fig 3 reveal that the proposed Mamba-based policy consistently outperforms all baselines in overall performance. On the one hand, the Mamba-based policy attains the highest converged return, surpassing not only the MLP-based and LSTM-based architectures but also the previously SOTA Transformer-based policy. This outcome highlights the superior capability of the Mamba-based architecture to capture long-horizon temporal dependencies from high-dimensional observations, thereby facilitating the exploration of more effective control policies. On the other hand, the Mamba-based policy exhibits substantially lower variance throughout the training process, indicating more stable skill acquisition and consistent

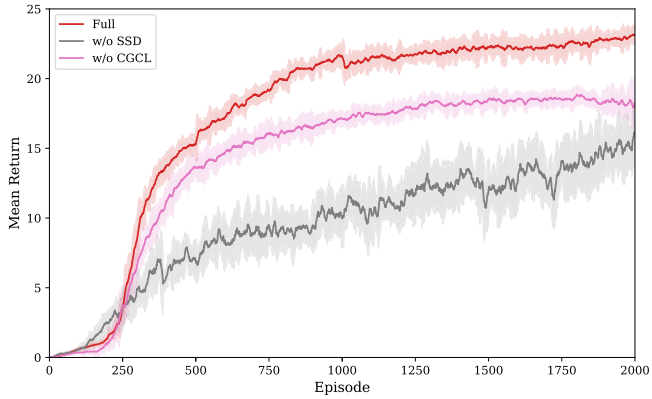

Fig. 4. Ablation study results demonstrating the contribution of key components. The comparative learning curves indicate that the removal of either the SSD module or the CGCL strategy leads to a significant decrease in the converged return, validating the necessity of each component for the overall performance.

performance, which in turn reflects enhanced robustness against minor perturbations. Notably, the MLP-based policy learns quickly in the initial stages due to its structural simplicity but ultimately converges to a low return, as its limited capacity prevents effective modeling of long-term temporal dependencies. Overall, these comparative experiments provide compelling evidence for the superiority of the proposed Mamba-based architecture in terms of both performance and robustness.

B. Ablation Studies

To further evaluate the contribution of individual components within the proposed method to overall locomotion performance, a series of ablation experiments is conducted. Initially, the SSD module, which serves as the core component for temporal information processing in Mamba-2, is removed while keeping all other components unchanged. The corresponding learning curve is shown in Fig 4. It can be observed that eliminating the SSD module leads to a substantial decrease in the average return, underscoring its pivotal role in enabling the model to perform content-based reasoning over sequential data. Interestingly, during the early training stages, the ablated model without SSD achieved higher returns compared to the full Mamba-2. This behavior can be attributed to the notion that the reduced architectural complexity facilitates gradient propagation and optimization, allowing the network to quickly fit to short-term and local sequential patterns. However, such a simplified model tends to adopt reactive policies that heavily rely on immediate state information, lacking the ability to capture long-range dependencies, which ultimately leads to inferior long-horizon performance and unstable convergence. These observations collectively indicate that the SSD module not only ensures superior asymptotic performance but also enhances system stability throughout the training process, making it an essential component for achieving robust and high-quality locomotion. Furthermore, to strictly quantify the impact of the curriculum strategy, the learning curve corresponding to the training process without

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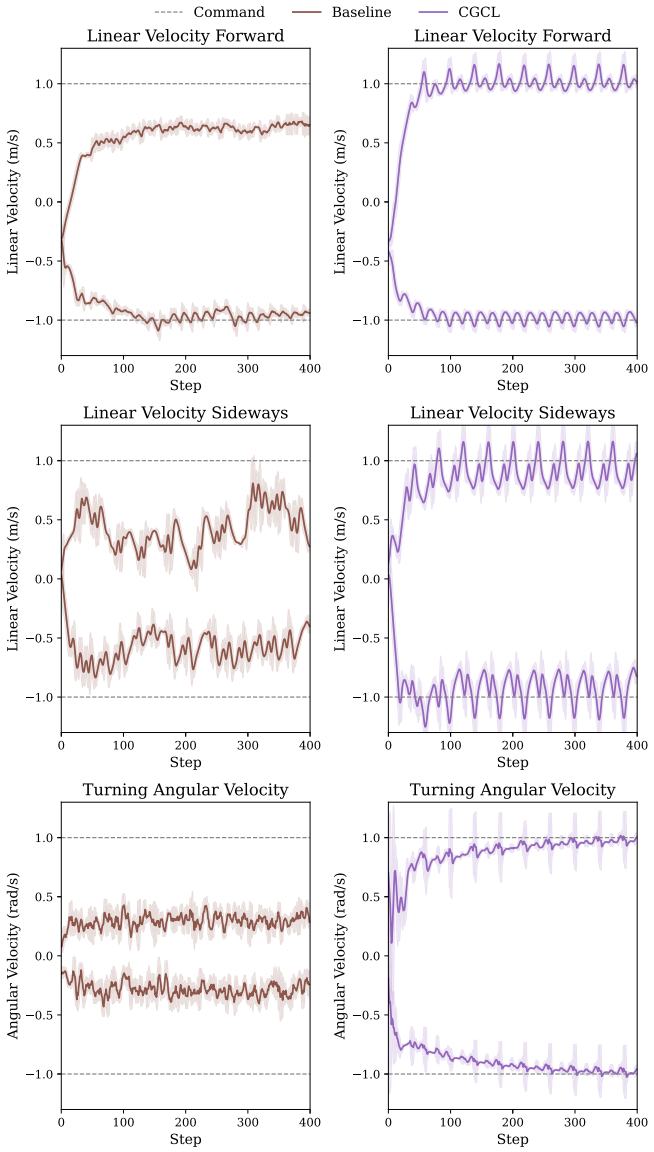

Fig. 5. The CGCL method enables accurate tracking of velocity commands across all three motion dimensions.

CGCL is also incorporated into Fig 4. As illustrated, the absence of the CGCL strategy results in a marked degradation in the final converged return. This performance gap can be attributed to the fact that without the progressive guidance of the curriculum, the agent struggles to escape local optima characterized by conservative, low-speed behaviors, thereby failing to maximize the velocity tracking rewards. Consequently, this comparison quantitatively confirms that the proposed CGCL method is instrumental in enabling the agent to attain a significantly higher level of overall locomotion performance.

Subsequently, an ablation experiment is conducted to further assess whether the proposed CGCL method enhances velocity tracking performance in humanoid locomotion. In this experiment, the baseline is defined by randomly sampling the forward linear velocity in the range of $[-1, 1]$ m/s, sideways linear velocity in $[-1, 1]$ m/s, and turning angular velocity in $[-1, 1]$ rad/s, which is the commonly used approach in training

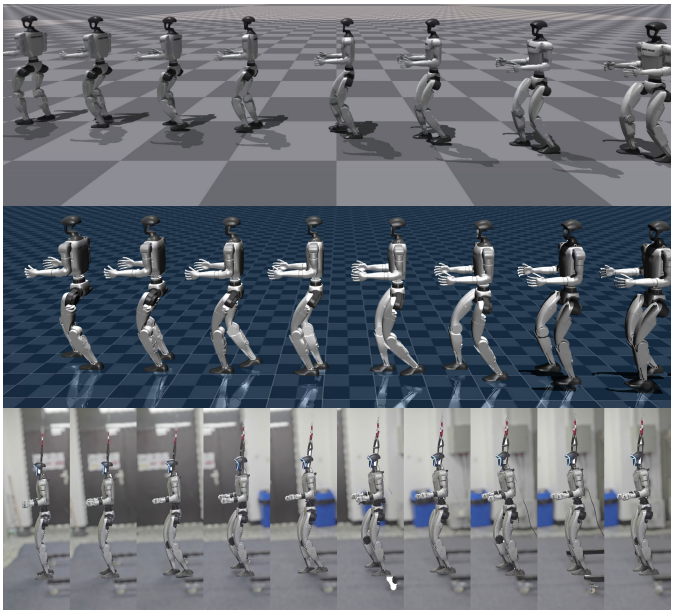

Fig. 6. Demonstration of sim-to-sim and sim-to-real transfer.

settings. The velocity commands for the trained policy are set to ± 1 m/s for forward linear velocity, ± 1 m/s for sideways linear velocity, and ± 1 rad/s for turning angular velocity, and the results are shown in Fig 5. It can be observed that the baseline, without CGCL, exhibits a steady-state error in tracking the forward linear velocity, significant fluctuations and errors when tracking sideways linear velocity, and is unable to track the turning angular velocity. In contrast, the CGCL method proposed in this paper achieves excellent tracking performance across all three motion dimensions, particularly demonstrating a significant improvement in tracking angular velocity.

C. Simulation-to-Simulation and Simulation-to-Reality

The policy trained in Isaac Gym is initially subjected to a sim-to-sim transfer, which serves to evaluate its generalization capability across different simulation environments. This process ensures that the policy can operate reliably under diverse simulated conditions, thereby providing a more robust foundation for subsequent deployment from simulation to the real world. The results of transferring the policy to MuJoCo are shown in Fig 6. As a high-fidelity physics engine [33], MuJoCo provides a stringent test environment. Without any simulation-specific fine-tuning, the policy trained with the proposed method is directly transferred to MuJoCo, where it demonstrates stable and robust locomotion. These findings highlight the strong generalization ability of the approach, confirming that its effectiveness is not contingent on the characteristics of a particular simulator.

Subsequently, the trained policy is directly deployed on the physical humanoid robot, as illustrated in Fig 6. To bridge the gap between simulation and reality, extensive domain randomization of dynamic parameters is applied during training. Specifically, additive noise is introduced to the observations, including angular velocity, joint positions, joint velocities, and projected gravity, in order to emulate sensor noise encountered

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in real-world scenarios. Moreover, noise is applied to motor strength, motor offset, PD gains, and payloads to capture actuator deviations and potential malfunctions. In addition, random system delays are incorporated to account for communication latency in physical systems. When the policy trained with domain randomization is deployed on the real humanoid robot, it demonstrates stable forward locomotion and accurate command following. The only observable differences are a slightly prolonged support phase and a marginally reduced push-off force, which are consistent with motor bandwidth limitations and unmodeled compliance characteristics. Furthermore, to verify the robustness against external interference in the physical environment, external push forces are applied to the robot during locomotion. It is observed that the proposed policy exhibits rapid reflex responses, effectively counteracting perturbations by adjusting foot placement and body posture to resume stable velocity tracking. This robustness is attributed to the domain randomization of external push forces incorporated during simulation training, enabling the learned policy to adapt to external disturbances. These results collectively demonstrate that the proposed method enables reliable transfer of the learned policy from simulation to reality, thereby validating its effectiveness for real-world humanoid locomotion.

V. Conclusion

In this paper, a novel curriculum-enhanced hierarchical RL framework is proposed for training humanoid locomotion policies. Drawing inspiration from the demonstrated success of Mamba in LLMs and computer vision, Mamba-2 is employed as the backbone of the Actor-Critic architecture to perform temporal modeling of historical observations, achieving superior performance over prior Transformer-based SOTA approaches. In addition, a CGCL method is proposed to mitigate the common tendency of humanoid robots to prioritize low-speed motion at the expense of velocity command tracking, thereby enhancing their capacity to accurately follow velocity commands. Extensive comparative and ablation experiments validate the effectiveness of the proposed framework, while real-world deployment further corroborates its robustness and stability.

Nevertheless, the current method relies solely on proprioceptive information, whereas intelligent autonomy in humanoid robots fundamentally requires external perception. Future research will focus on developing humanoid locomotion policies that integrate external sensory modalities.

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